

# Exercise Session 2

## MDMC Spring 2025

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## Exercise General Information

- Interviews
  - 10 minutes discussion to test your understanding of the exercise
  - Good occasion to discuss your doubts and questions
- Report feedback
  - After your interview, you will get a detailed feedback via Moodle
    - No grade
    - Overall comment and detailed correction of the exercises

**Exercises contribute to 1/2 of final grade! We count the best 5 out of the 6 reports for your exercise grade.**

## Notebooks Reminder

- Always access the notebooks via the rocket button on the top right of the code files and choose JupyterHub to launch [noto.epfl.ch](https://noto.epfl.ch)
- **Make sure to access noto this way each time you begin the exercise to ensure you have the latest version!**



# Exercise Structure



## Learning goals

Follow a (re)introduction to statistical mechanics

Learn differences between canonical and microcanonical ensembles

Understand how to calculate the Boltzmann distribution for a harmonic oscillator



## Chapter in script

Chapter 2 - Statistical Mechanics in a Nutshell



## Resources

Understanding Molecular Simulation, Frenkel & Smit, p.9-17

## Exercise 2 - Intro & Tips

Today we'll be building a tool to compute the probability distribution of the energy of a system using Boltzmann statistics. The focus of the exercise is to refresh your notions of statistical mechanics, which is the foundation for all simulation techniques we'll see during the course.

- The theoretical part is about statistical thermodynamics.

Be sure to know what we mean by

- Thermodynamic ensembles
- Microstate
- Partition function and why it's useful
- Boltzmann distribution

## Exercise 2 - Intro & Tips

- The practical part is about computing the Boltzmann distribution of a fictitious harmonic oscillator.

For simplicity

- Boltzmann constant = 1
- $\text{Beta} = 1/T$
- Energies represented as integer values

## Exercise 2 - Intro & Tips

- Modify the code below to translate into code concepts from statistical mechanics

```
# MODIFY HERE
# set number of energy levels and temperatures here
n_energy_levels= 0
reducedTemperatures=[0, 0, 0, 0]

def calculateStateOccupancy(T, i):
    # No degeneracy
    return 1 # MODIFY HERE

def calculateStateOccupancy_s1(T, i):
    # Degeneracy s+1
    return 1 # MODIFY HERE

def calculateStateOccupancy_s2(T, i):
    # Degeneracy s+2
    return 1 # MODIFY HERE

functions ={
    "no_degeneracy": calculateStateOccupancy,
    "s+1": calculateStateOccupancy_s1,
    "s+2": calculateStateOccupancy_s2
}

for f in functions.keys():
    fig, ax =plt.subplots(1)
    ax.set_title(f)
    ax.set_xlabel("Energy level")
    ax.set_ylabel("Occupancy")

    calculateOccupancy=functions[f]

    for reducedTemperature in reducedTemperatures:
        distribution = [] # For each state there is one entry
        partition_function=np.float64(0.0)

        for i in range(n_energy_levels):
            stateOccupancy=calculateOccupancy(reducedTemperature, i)
            distribution.append(1.0) # MODIFY HERE
            partition_function+=1.0 # MODIFY HERE

        ax.plot(distribution/partition_function, label=reducedTemperature)

ax.legend()
plt.show()
```

## Exercise 2 - Intro & Tips

- Remember there is also Exercise 8 (often accidentally skipped in previous years!)

```
# INSERT YOUR WORKING CODE HERE
# ADAPT it for the linear rotor case
# defining a new calculateStateOccupancy_rotor function and a "linear rotor" key

# Then run the following cell (after having edited it where indicated!)
```

```
for f in functions.keys():
    fig, ax = plt.subplots(1)
    ax.set_title(f)
    ax.set_xlabel("Reduced Temperature")
    ax.set_ylabel("Partition Function")

    calculateOccupancy=functions[f]
    Z = [] # This variable will store the partition function
          # for different reduced temperatures

    for reducedTemperature in reducedTemperatures:
        distribution = [] # For each state there is one entry
        partition_function=np.float64(0.0)

        for i in range(n_energy_levels):
            stateOccupancy=calculateOccupancy(reducedTemperature, i)
            distribution.append(1.0) # MODIFY HERE
            partition_function+=1.0 # MODIFY HERE
        Z.append(1.0) # MODIFY HERE
    ax.plot(reducedTemperatures, Z, label="Z Exact")
    ax.plot(reducedTemperatures, Z, label="Z Approximated") # MODIFY HERE
                                                         # insert approximated Z

ax.legend()
plt.show()
```